

23 Quantum channel noise and channel capacity

This chapter introduces the notion of *noisy quantum channels*, and the different types of “quantum noise” that affect qubit messages passed through such channels. The main types of noisy channel reviewed here are the *depolarizing*, *bit-flip*, *phase-flip*, and *bit-phase-flip* channels. Then the *quantum channel capacity* χ is defined through the *Holevo–Schumacher–Westmoreland* (HSW) theorem. Such a theorem can conceptually be viewed as the elegant quantum counterpart of *Shannon’s (noisy) channel coding theorem*, which was described in Chapter 13. Here, I shall not venture into the complex proof of the HSW theorem but only provide a background illustrating the similarity with its classical counterpart. The resemblance with the channel capacity χ and the *Holevo bound*, as described in Chapter 21, and with the classical *mutual information* $H(X; Y)$, as described in Chapter 5, are both discussed. For advanced reference, a hint is provided as to the meaning of the still not fully explored concept of *quantum coherent information*. Several examples of quantum channel capacity, derived from direct applications of the HSW theorem, along with the solution of the maximization problem, are provided.

23.1 Noisy quantum channels

The notion of “noisiness” in a classical communication channel was first introduced in Chapter 12, when describing *channel entropy*. Such a channel can be viewed schematically as a probabilistic relation between two random sources, X for the originator, and Y for the recipient. These sources are defined by symbol alphabets, $X = \{x_1, x_2\}$ and $Y = \{y_1, y_2\}$ for instance, and their associated probabilities $p(x_i)$, $p(y_j)$. An originator message is, thus, a string of n symbols $x_i \in X$ randomly selected from X according to the distribution $p(x_i)$, which is transformed at the recipient’s end into a string of n symbols $y_i \in Y$ randomly selected from Y according to the distribution $p(y_i)$. An ideal or noiseless channel is such that the knowledge of any symbol x_i input from the originator absolutely conditions the knowledge of the symbol y_j obtained at the recipient’s end. Thus, for any originator symbol x_i , there must be a recipient symbol y_j , such that $p(y_j|x_i) = 1$ and $p(y_j|x_{i \neq i}) = 0$. The set of conditional probabilities $p(y_j|x_i)$ can be put in the form of a *transition matrix* $P(Y|X)$ and in the case of a noiseless channel with

a two-symbol alphabet (binary channel) we must have

$$P(Y|X) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \text{ or } \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (23.1)$$

In the nonideal case where the channel is corrupted by noise, we have $p(y_j|x_i) = 1 - \varepsilon$, where $0 < \varepsilon < 1$ defines the *symbol error probability* (the probability of mistaking one symbol for another). The error probability can be the same for the two symbols (binary symmetric channel), which gives, for instance,

$$P(Y|X) = \begin{pmatrix} 1 - \varepsilon & \varepsilon \\ \varepsilon & 1 - \varepsilon \end{pmatrix}. \quad (23.2)$$

The above recall from the classical information theory will now help to conceive of the effect of noise in a *quantum* communication channel.

As we have seen in Chapter 22, a quantum message M is a “block” represented by the tensor state $|M\rangle = |q_1 q_2 \dots q_n\rangle$, with each qubit $|q_i\rangle$ being randomly selected from a given symbol alphabet $\{|x_k\rangle\}_{k=1\dots N}$ of size N . Given the fact that each symbol $|x_k\rangle$ is associated with an occurrence probability p_k , it is possible to define a *symbol density operator* according to:

$$\rho = \sum_{k=1}^N p_k |a_k\rangle \langle a_k|. \quad (23.3)$$

Consistently, the full symbol block of length n is characterized by the *density operator*:

$$\rho_M = \rho \otimes \rho \otimes \dots \otimes \rho \equiv \rho^{\otimes n}, \quad (23.4)$$

which represents the originator’s message M . As with the definition of a density operator, we have $\text{tr}(\rho_M) = 1$. Consequently, some message σ_M may be received at the recipient’s end. The correspondence between the originator’s and the recipient’s messages may be defined through a transformation, or *quantum operation*, ε , such that

$$\sigma_M = \varepsilon(\rho_M). \quad (23.5)$$

Such a quantum operation ε must be trace preserving, so that σ_M is a density operator with $\text{tr}(\sigma_M) = 1$. Since the message transformation from originator to recipient is fully defined by the operation ε , it is customary to refer to ε as the *quantum channel* itself.

In the ideal or “noiseless” case, one would expect that the quantum channel ε corresponds to some unitary (trace-preserving) transformation U , which gives

$$\sigma_M = \varepsilon(\rho_M) = U^+ \rho_M U. \quad (23.6)$$

Thus, according to the above operation the message is not invariant by transmission through the quantum channel ($\sigma_M \neq \rho_M$), but its integrity is fully conserved. Indeed, it only takes the recipient to apply the inverse operation ε^{-1} defined by $U^{-1} = U^+$ to the received message σ_M , to obtain $\sigma'_M = \varepsilon^{-1}(\sigma_M) = U \sigma_M U^+ \equiv \rho_M$ and, hence, fully retrieve the originator’s message. A trivial case of ideal or noiseless channel is given by $U = I^{\otimes n}$, where I is the 2×2 identity matrix. A more general definition for any

quantum channel ε is

$$\varepsilon(\rho_M) = \sum_k U_k^+ \rho_M U_k. \quad (23.7)$$

The definition in Eq. (23.7) is referred to as the *operator-sum representation* of the channel, with U_k being the *channel-operator elements*.

Consider next the case of *nonideal*, or *noisy quantum channels*. Clearly, such channels can be modeled by a quantum operation $\varepsilon(\rho_M)$, whose outcome σ_M must be *random*. Here, I shall describe four basic types of noisy quantum channels, assuming that the message is made of a single qubit.

Depolarizing channel

The first type of noisy quantum channel, called a *depolarizing channel*, is defined as follows:

$$\varepsilon(\rho) = p \frac{I}{2} + (1 - p)\rho, \quad (23.8)$$

where p is some probability distribution and I is the identity matrix. According to this definition, there is a probability $1 - p$ that the originator message ρ is left invariant by the channel, and a probability p that it is transformed into the message $\sigma = I/2$. Thus, the messages $\rho' = |0\rangle\langle 0|$ and $\rho'' = |1\rangle\langle 1|$, corresponding to the pure basis states $|0\rangle$ and $|1\rangle$, are transformed into

$$\begin{cases} \sigma' = \varepsilon(\rho') = \frac{p}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + (1 - p) \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 - \frac{p}{2} & 0 \\ 0 & \frac{p}{2} \end{pmatrix} \\ \sigma'' = \varepsilon(\rho'') = \frac{p}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + (1 - p) \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \frac{p}{2} & 0 \\ 0 & 1 - \frac{p}{2} \end{pmatrix}. \end{cases} \quad (23.9)$$

The above-defined quantum operation ε is referred to as the “depolarizing” channel, because it transforms a pure basis state ρ' , ρ'' into a uniformly mixed state $\sigma = I/2$ with a certain probability p . From the above results, it would appear that the operation $\varepsilon(\rho)$ is in fact equivalent to the following pure-state transformation:

$$\begin{cases} |0\rangle \rightarrow \sqrt{1 - \frac{p}{2}}|0\rangle \pm \sqrt{\frac{p}{2}}|1\rangle \\ |1\rangle \rightarrow \sqrt{\frac{p}{2}}|0\rangle \pm \sqrt{1 - \frac{p}{2}}|1\rangle. \end{cases} \quad (23.10)$$

However, in view of the recipient's qubit definitions in Eq. (23.10), such an interpretation is ambiguous, since the relative phases ($\pm$) between the $|0\rangle$ and $|1\rangle$ qubits are undefined. Then which are the actual recipient qubits? The answer can be obtained by analyzing the quantum operation defined in Eq. (23.8). We observe that there is a probability $\varepsilon = 1 - p$ that the communication channel accurately transmits the original message ρ . Thus, ε represents the probability of the recipient obtaining *nonerrored states*

$|q\rangle_{\text{in}} = |q\rangle_{\text{out}} = |0\rangle, |1\rangle$, corresponding to original states. The probability that the recipient states are errored is p . In this event, the communication channel “ignores” ρ and “chooses” instead to transmit $\sigma = I/2$, which is associated with the uniform mixed states $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$. In this event, the channel randomly outputs the errored states $|q\rangle_{\text{out}} = |+\rangle, |-\rangle$, regardless of the input states $|q\rangle_{\text{in}} = |0\rangle, |1\rangle$. The conclusion is that *there is a probability p that there is no mathematically defined relation between the channel input ($|q\rangle_{\text{in}}$) and output ($|q\rangle_{\text{out}}$) qubits*. Such an event corresponds to a *loss of coherence*, a unique quantum effect referred to as *decoherence*. It is beyond the scope of this book to analyze further the meaning and implication of quantum decoherence, as introduced by random channel noise. The only (but important!) lesson to be learnt here is twofold: (a) the quantum-channel transformation is accurately defined by the density matrix operation $\sigma = \varepsilon(\rho)$; (b) without supplemental information, the output states of the quantum channel cannot be accurately or predictably retrieved, because of the decoherence effect. However, it is possible to implement *quantum error correction*, and hence retrieve the input quantum message (ρ) and associated qubit symbols ($|q\rangle_{\text{in}}$), as will be described in Chapter 24.

Let us now look at the VN entropy change through the depolarizing channel. For any pure-state message ρ , we have, by definition, $S(\rho) = 0$ (there is no uncertainty as to what message is being sent). From the recipient’s end, and based on the result in Eq. (23.9), we have

$$\begin{aligned} S(\sigma) &= S(\sigma') = S(\sigma'') = -\text{tr}(\sigma \log \sigma) \\ &= \frac{p}{2} \log \frac{p}{2} - \left(1 - \frac{p}{2}\right) \log \left(1 - \frac{p}{2}\right) \equiv f\left(\frac{p}{2}\right). \end{aligned} \quad (23.11)$$

Thus, the quantum channel has an uncertainty of the amount $S(\sigma) = f(p/2)$, which as we know from earlier chapters is maximal for the argument $p/2 = 1/2$ or $p = 1$, which gives $S(\sigma) = 1$ bit. Clearly, this limiting case corresponds to a *useless quantum channel*, where the received message is deterministically $\sigma = I/2$ and, no matter what the original message information is, the net uncertainty is one classical bit!

To conclude this example formally, it is left as a basic exercise to show that the depolarizing channel can be put in the operator-sum representation

$$\begin{aligned} \varepsilon(\rho) &= p \frac{I}{2} + (1 - p)\rho \\ &\equiv \frac{q}{3} (X\rho X + Y\rho Y + Z\rho Z) + (1 - q)\rho, \end{aligned} \quad (23.12)$$

where $q = 3p/4$ and X, Y, Z are the *Pauli matrices* (see Chapter 16). This expression, which also holds in the general case where ρ is nondiagonal, shows that the effect of noise is to transform with equal probability $q/3$ the originator’s message ρ into either $X\rho X$, or $Y\rho Y$ or $Z\rho Z$, and to leave it invariant with probability $1 - q$.

Bit-flip channel

As a second type of noisy quantum operation, I introduce next the *bit-flip* channel, as defined in the operator-sum representation:

$$\begin{aligned}\varepsilon(\rho) &= pX\rho X + (1-p)\rho \\ &\equiv U_0\rho U_0 + U_1\rho U_1,\end{aligned}\quad (23.13)$$

where, as usual, p is some probability distribution, and where we have the following channel-operator elements:

$$\begin{cases} U_0 = \sqrt{p}X = \sqrt{p}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ U_1 = \sqrt{1-p}I = \sqrt{1-p}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \end{cases}\quad (23.14)$$

It is straightforward to establish that the action of the first operation term $X\rho X$ in Eq. (23.13) is to transform the originator's message ρ into the recipient message $\sigma = \varepsilon(\rho)$ with the flipped coefficients $\sigma_{11} = \rho_{22}$ and $\sigma_{22} = \rho_{11}$. Such a transformation has probability p of occurring, while the originator's message is left unchanged ($\sigma = \rho$) with probability $1-p$. Effecting the calculation with the qubit message $|q\rangle = \alpha|0\rangle + \beta|1\rangle$ ($\alpha = \sqrt{\rho_{11}}$, $\beta = \sqrt{\rho_{22}}$), it is also straightforward to obtain the channel transformation:

$$|q\rangle \rightarrow |q'\rangle = \sqrt{(1-p)|\alpha|^2 + p|\beta|^2}|0\rangle + \sqrt{p|\alpha|^2 + (1-p)|\beta|^2}|1\rangle. \quad (23.15)$$

Thus, if the originator's message is the pure-state qubit $|0\rangle$ (i.e., $\alpha = 1$, $\beta = 0$), the recipient receives the message

$$|q'\rangle = \sqrt{1-p}|0\rangle + \sqrt{p}|1\rangle. \quad (23.16)$$

Clearly, the symbol-error (or cbit-error) probability is $\varepsilon = p$, meaning that the channel has the probability p of outputting the errored message $|1\rangle$ when $|0\rangle$ was sent. The same conclusion applies in the opposite case, which illustrates the notion of *bit flipping* as a noise effect in quantum channels. It is left as an interesting exercise to analyze the effect of bit-flip transformation in the case where ρ is nondiagonal. As expected, the outcome of the exercise is that an originator's qubit message $|q\rangle = \alpha|0\rangle + \beta|1\rangle$ is "flipped" through the noisy channel into a recipient's message $|q\rangle = \beta|0\rangle + \alpha|1\rangle$ under an outcome probability p . Finally, the entropy difference between a pure-state originator's message $\rho = |0\rangle\langle 0|$ and the recipient's message $\sigma = (1-p)|0\rangle\langle 0| + p|1\rangle\langle 1|$ is

$$\begin{aligned}\Delta S &= S(\sigma) - S(\rho) = [-(1-p)\log(1-p) - p\log p] - 0 \\ &\equiv f(1-p) = f(p),\end{aligned}\quad (23.17)$$

with the same result being obtained with $\rho = |1\rangle\langle 1|$. The maximum uncertainty ($\Delta S = 1$ or one cbit), corresponds to the case $p = 0.5$, meaning that, according to Eq. (23.13), there is a 50% chance that the qubit amplitudes are flipped when passing through the channel. Interestingly, there is no entropy change ($\Delta S = 0$) if the qubit amplitudes are deterministically flipped ($p = 1$). This is because there is no difference in information

contents if the cbit polarity (the definition of 0 and 1, or $|0\rangle$ and $|1\rangle$) is *systematically* flipped.

Phase-flip channel

As a third example of noisy quantum operation, I define next the *phase-flip* channel in the operator-sum representation:

$$\varepsilon(\rho) = pZ\rho Z + (1-p)\rho \equiv U_0\rho U_0 + U_1\rho U_1, \quad (23.18)$$

with the corresponding elements:

$$\begin{cases} U_0 = \sqrt{p}Z = \sqrt{p} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \\ U_1 = \sqrt{1-p}I = \sqrt{1-p} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \end{cases} \quad (23.19)$$

It is again left as an exercise to show that the originator's message $|q\rangle = \alpha|0\rangle + \beta|1\rangle$ is transformed into $|q'\rangle = \alpha|0\rangle - \beta|1\rangle$ with probability p and is left invariant otherwise. Thus, the action of the noisy channel is to flip (with probability p) the relative phase $e^{i\delta}$ randomly between the complex amplitudes α, β from e^{i0} to $e^{i\pi}$. In the case of qubit states $|q\rangle = |0\rangle, |1\rangle$ it is found that the entropy change is zero ($\Delta S = 0$), clearly because no phase change can affect a single qubit. In the case $|q\rangle = |+\rangle$, or $\rho = |+\rangle\langle+|$, one finds the recipient's message:

$$\sigma = p|-\rangle\langle-| + (1-p)|+\rangle\langle+|, \quad (23.20)$$

which in the basis $\{|+\rangle, |-\rangle\}$ is diagonal, i.e.,

$$\sigma = \begin{pmatrix} p & 0 \\ 0 & 1-p \end{pmatrix}, \quad (23.21)$$

hence, the net entropy difference $\Delta S = S(\sigma) - S(\rho) = f(p) - 0 \equiv f(p)$, with the same result being obtained with $\rho = |-\rangle\langle-|$. The maximum uncertainty ($\Delta S = 1$ or one cbit), corresponds to the case $p = 0.5$, meaning that, according to Eq. (23.18), there is a 50% chance that the relative qubit phase is flipped by $e^{i\pi}$ when passing through the channel. Interestingly, there is no entropy difference ($\Delta S = 0$) if the phase is deterministically changed ($p = 1$). This is because there is no difference in information contents if the cbit polarity (the definition of 0 and 1 or $|+\rangle$ and $|-\rangle$) is systematically flipped.

Bit-phase-flip channel

As a fourth example of noisy quantum operation, we define next the *bit-phase-flip* channel in the operator-sum representation:

$$\begin{aligned} \varepsilon(\rho) &= pY\rho Y + (1-p)\rho \\ &\equiv U_0\rho U_0 + U_1\rho U_1, \end{aligned} \quad (23.22)$$

with the corresponding elements:

$$\begin{cases} U_0 = \sqrt{p}Y = \sqrt{p} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \\ U_1 = \sqrt{1-p}I = \sqrt{1-p} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \end{cases} \quad (23.23)$$

It is again left as an exercise to show that the originator message $|q\rangle = \alpha|0\rangle + \beta|1\rangle$ is transformed into $|q'\rangle = \beta|0\rangle - \alpha|1\rangle$ with probability p and is left invariant otherwise. The action of the noisy channel is to flip both the phase and the complex amplitudes. It is easily found that for any originator messages $\rho = |0\rangle\langle 0|$, $\rho = |1\rangle\langle 1|$, $\rho = |+\rangle\langle +|$ or $\rho = |-\rangle\langle -|$, the net entropy difference is $\Delta S = S(\sigma) - S(\rho) = f(p)$. It is clear that for $\rho = |0\rangle\langle 0|$, $\rho = |1\rangle\langle 1|$ this difference is only attributable to the bit-flipping effect, while for $\rho = |+\rangle\langle +|$ or $\rho = |-\rangle\langle -|$ it is attributable to the phase-flipping effect.¹

Other noise processes, which are beyond the scope of these chapters but very important in the analysis of time evolution of qubits in quantum channels, concern *amplitude damping* and *phase damping*.

23.2 The Holevo–Schumacher–Westmoreland capacity theorem

In this section, I shall introduce the concept of *quantum channel capacity*, based on the *Holevo–Schumacher–Westmoreland* (HSW) theorem. The HSW theorem defines the channel capacity χ under which, at a rate $R < \chi$, it is possible to transmit quantum messages with arbitrary small probability of error or information loss. It can be viewed conceptually as the elegant quantum counterpart of *Shannon’s channel coding theorem*, which was described in Chapter 13. Here, the term “capacity” refers, in fact, to the same concept as in *classical information*, while assuming that such information is encoded in quantum messages that are passed through noisy quantum channels.

The formal demonstration of the HSW theorem being particularly involved and tedious, it will not be considered here. For the general interest and purpose of this chapter, it will suffice to state the HSW theorem “as is:” given a quantum-symbol source $\{\rho_i\}$ with associated probability distribution $\{p_i = p(\rho_i)\}$, the capacity χ of a noisy quantum channel ε providing the operation $\sigma_i = \varepsilon(\rho_i)$ is given by the maximum difference:

$$\begin{aligned} \chi &= \max[S\langle\sigma\rangle - \langle S(\sigma)\rangle] \\ &= \max_{\{p_i\}} \{S[\varepsilon(\langle\rho_i\rangle)] - \langle S[\varepsilon(\rho_i)]\rangle\}, \end{aligned} \quad (23.24)$$

¹ Flipping complex amplitudes leave $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ invariant, while resulting in a global phase factor $e^{i\pi} = -1$, for $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$, which does not modify the definition of $|-\rangle$.

where the brackets $\langle \rangle$ indicate an averaging over the distribution $\{p_i\}$. This definition can be put into the more explicit or complete formulation:

$$\begin{aligned}\chi &= \max_{\{p_i\}} \left[S \left(\sum_i p_i \sigma_i \right) - \sum_i p_i S(\sigma_i) \right] \\ &= \max_{\{p_i\}} \left\{ S \left[\varepsilon \left(\sum_i p_i \rho_i \right) \right] - \sum_i p_i S[\varepsilon(\rho_i)] \right\}.\end{aligned}\quad (23.25)$$

As with Shannon's channel coding theorem, the capacity χ represents the maximum code rate for which the probability of transmission error can be made arbitrarily small, assuming sufficiently long message lengths. The following provides a background and a basic interpretation for the HSW theorem, showing how it nicely parallels Shannon's channel coding theorem.

As we have seen in Chapter 22, a quantum message of length n can be conceived as an n -qubit "block" represented by the tensor state $|M\rangle = |x_1 x_2 \dots x_n\rangle$, with each qubit $|x_k\rangle$ being randomly selected with an occurrence probability p_i , from a given symbol alphabet $\{|a_i\rangle\}_{i=1\dots N}$ of size N . We assume that the originator chooses among 2^{nR} message possibilities, where $R > 0$. This limited set of "quantum codewords" of length n can be referred to as a "code," and R as a *code rate*. A *symbol density operator* ρ_{m_k} can be associated with any message qubit in position k , where m_k is an index selected from the alphabet $i = 1, 2, \dots, N$. Hence, the originator's message block or codeword is characterized by a density operator ρ_M constructed from the n -tensor product:²

$$\rho_M = \rho_{m_1} \otimes \rho_{m_2} \otimes \dots \otimes \rho_{m_n}. \quad (23.26)$$

We note that since the message block is defined as an n -tensor product, there is no entanglement between any of the qubits therein. The recipient message, or codeword is received as

$$\sigma_M = \sigma_{m_1} \otimes \sigma_{m_2} \otimes \dots \otimes \sigma_{m_n}, \quad (23.27)$$

where

$$\sigma_{m_k} = \varepsilon(\rho_{m_k}). \quad (23.28)$$

Following the description in Chapter 22 (but not being concerned here about compression), it is clear that a specific codeword σ_M corresponds to a set of $N' = N^n$ eigenvalues $\Lambda_{MK} = \lambda_{m_1 k_1} \lambda_{m_1 k_2} \dots \lambda_{m_1 k_{n_1}}$ and eigenvectors $|\Lambda_{MK}\rangle = |\lambda_{m_1 k_1} \lambda_{m_1 k_2} \dots \lambda_{m_1 k_{n_1}}\rangle$, such that σ_M has the diagonal form

$$\begin{aligned}\sigma_M &= \sum_K \Lambda_{MK} |\Lambda_{MK}\rangle \langle \Lambda_{MK}| \\ &= \begin{pmatrix} \Lambda_{M1} & 0 & \dots & 0 \\ 0 & \Lambda_{M2} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \Lambda_{MN'} \end{pmatrix},\end{aligned}\quad (23.29)$$

² For instance, $|M\rangle = |a_1 a_2 a_3\rangle$ is represented by the density operator $\rho_M = \rho_1 \otimes \rho_2 \otimes \rho_2 \otimes \rho_3$.

with $\sum_K \Lambda_{MK} = 1$. Next, define $\langle S \rangle = \langle S(\sigma) \rangle$ as the mean value of the per-symbol VN entropy of the received codeword, according to:

$$\langle S \rangle = \sum_i p_i S(\sigma_i). \quad (23.30)$$

Given any parameter $\varepsilon > 0$, and based on the concept described in Chapter 22, we can define the ε -typical subspace Ω for σ_M , based on the parameters Λ_{MK} , $\langle S \rangle$ and spanned by the eigenvectors $|\Lambda_{MK}\rangle$ according to the condition:

$$\left| \frac{1}{n} \log \frac{1}{\Lambda_{MK}} - \langle S \rangle \right| \leq \varepsilon. \quad (23.31)$$

Let P_M be the projector onto this ε -typical subspace Ω . Based on Eq. (22.30), we know that the dimension of this subspace, $\text{tr}(P_M)$, is upper-bounded according to

$$\text{tr}(P_M) \leq 2^{n(\langle S \rangle + \varepsilon)}. \quad (23.32)$$

We also know that for any σ_M with sufficiently long length n and for any δ such that $0 < \delta < 1$, the probability $p = p(\sigma_M \in \Omega)$ of projecting onto the typical subspace is lower-bounded according to:

$$p = \langle \text{tr}(\sigma_M P_M) \rangle \geq 1 - \delta = p_{\min}, \quad (23.33)$$

where the brackets have the meaning of an *expectation value*, i.e., the projection probability as averaged over all possible message codewords σ_M . This result shows that the probability that any received codeword σ_M belongs to the typical subspace has a lower bound $p_{\min}$, which is nonzero. I shall now introduce the notion of *error probability*, which is the probability that a POVM measurement of σ_M by the recipient fails to identify σ_M as one of the codewords from the typical set Ω .

Consider, indeed, that given the typical set $\Omega = \{\sigma_M\}$ of size 2^{nR} , there is a corresponding set $\{E_M\}$ of *POVM measurement operators* (see Chapter 17) of the same size. The POVM set is completed with the measurement operator E_0 for all other possible codewords, defined as

$$E_0 = I^{\otimes n} - \sum_{M \neq 0} E_M. \quad (23.34)$$

The probability p_M of the recipient identifying a specific message M in Ω while using a POVM measurement E_M is given by

$$p_M = \text{tr}(\sigma_M E_M) \quad (23.35)$$

and in the opposite case, where such identification is unsuccessful, the error probability p_M^{error} is:

$$p_M^{\text{error}} = 1 - p_M = 1 - \text{tr}(\sigma_M E_M). \quad (23.36)$$

If we assume that the originator *uniformly* selects the messages from the codeword set $\{\rho_M\}$, it is then possible to define an *average error probability* $\bar{p}_{\text{error}}$ through

$$\bar{p}_{\text{error}} = \frac{1}{2^{nR}} \sum_M p_M^{\text{error}}. \quad (23.37)$$

Define $\langle \bar{p}_{\text{error}} \rangle$ as the expectation value of $\bar{p}_{\text{error}}$ over all possible codewords. The formal demonstration of the HSW theorem then leads to the following result, which I shall here provide “as is:”

$$\langle \bar{p}_{\text{error}} \rangle \leq 4\delta + (2^{nR} - 1)2^{-n(\chi - 2\epsilon)}, \quad (23.38)$$

with

$$\chi = S(\langle \sigma \rangle) - \langle S(\sigma) \rangle. \quad (23.39)$$

The result in Eq. (23.38) shows that provided the condition for the code rate R ,

$$R < S(\langle \sigma \rangle) - \langle S(\sigma) \rangle = \chi, \quad (23.40)$$

and for long message lengths ($n \rightarrow \infty$) the expected error probability is upper-bounded according to $\langle \bar{p}_{\text{error}} \rangle \leq \delta' = 4\delta$ with δ being *any* real such that $0 < \delta < 1$, i.e., which can be chosen arbitrarily close to zero. The maximization of χ , which yields the quantum channel *capacity*, is an issue at least as complex as in the case of classical communication channels. Suffice it here to infer that under the condition $R < \max(\chi)$ it is possible to find another class of quantum codes for which $\bar{p}_{\text{error}} < \delta''$, with δ'' being arbitrarily close to zero. The key conclusion is, therefore:

Given a quantum channel ε , there exist quantum codes with long codeword size n and code rate $R < \max(\chi)$, for which the error probability $\bar{p}_{\text{error}}$ (of the recipient failing to identify a codeword $\sigma_M = \varepsilon(\rho_M)$ given an originator codeword ρ_M) can be made arbitrarily small.

This conclusion defines the key condition under which “error-free” transmission of quantum messages through noisy quantum channels can be achieved. It is also possible to show the converse, namely: “for $R > \chi$ there exists no quantum code of any codeword size able to achieve error-free transmission,” meaning that $\bar{p}_{\text{error}}$ is irremediably bounded away from zero.

As stated earlier, there exists a nice parallel between the HSW “capacity” theorem and Shannon’s coding theorem for classical, noisy communication channels. To recall from Chapter 13, the classical channel capacity C is defined as the mutual information maximum:

$$C = \max_{p(x)} H(X; Y). \quad (23.41)$$

The coding theorem states that:

Given a noisy transmission channel, there exist binary codes of sufficiently long codeword size, with 2^{nR} codewords of sufficient length n , and code rate $R < C$, for which an originator message can be transmitted to a recipient with arbitrary low error probability.

It is clear that Shannon's coding theorem and the HSW theorem are conceptually very similar. As in QIT, the typical set of codewords (or of typical sequences) is defined under the condition (Eq. (13.25)):

$$\left| \frac{1}{n} \log \frac{1}{p(x)} - H(X) \right| \leq \varepsilon, \quad (23.42)$$

where $p(x)$ is the *codeword* probability distribution associated with the *symbol* source X , and the entropy $H(X)$. This condition parallels that in Eq. (23.29), with $p(x) \rightarrow \Lambda_{MK}$ and $H(X) \rightarrow \langle S \rangle$. In the classical case, the dimension of the typical set is upper-bounded by $2^{n(H+\varepsilon)}$, while in the quantum case, the bound is $2^{n(\langle S \rangle + \varepsilon)}$, see Eq. (23.30). Because of this analogy, it is clear that the HSW theorem owes a great deal to Shannon's classical analysis, even if its background assumptions are hardly reducible to any classical conception. Then we may ask: "Is quantum channel capacity in any way reducible to Shannon's original concept?" The following discussion, although with no pretence at providing any academic proof, may guide towards some form of a satisfactory answer; we expected no!

We may now conclude this section with a discussion about the connection between the HSW theorem and the *Holevo bound* described in Chapter 21, and clarifying the difference between χ and the classical channel capacity C . Indeed, letting $\chi' = S(\langle \sigma \rangle) - \langle S(\sigma) \rangle$, we recognize from Chapter 21 that χ' is exactly the definition of the *Holevo bound* for the quantum source σ , and the HSW theorem seems simply to state that the channel capacity is given by $\chi = \max \chi'$. Then, apart from the maximization issue, we may wonder what is conceptually new and useful in this HSW theorem that has not already been captured into the Holevo bound concept. The answer to this apparent paradox is quite simple. In Chapter 21, we have considered the action of encoding and decoding classical information into or from a suite of quantum symbols ρ_x from an alphabet $\{\rho_i\}$, to be conveyed through some ideal quantum channel. The operation of encoding is the "preparation" of the state ρ_x , which means that the originator sets up the quantum system in the state ρ_x . The operation of decoding is the "measurement" by the recipient of any received symbol ρ_y , using the adequate POVM set for the identification of each symbol possibility. The full encoding and decoding operation can be virtually reduced into a classical information channel that relates two random sources X and Y . As we have learnt, the mutual information $H(X; Y)$ is bounded by $H(X)$, just as in the purely classical case, but the quantum channel introduces the extra (Holevo) bound χ such that

$$H(X; Y) \leq \chi \leq H(X). \quad (23.43)$$

As we have also seen, if the symbols ρ_i have orthogonal support in $\{\rho_i\}$, meaning that different qubit symbols are necessarily orthogonal, then $\chi = H(X)$ and the Holevo bound is reduced to the originator source entropy, which obliterates any particularity of using a quantum channel as opposed to a purely classical one.

Consider next the case of the noisy quantum channel. The encoding and decoding operations are conceptually the same as previously, except that decoding (POVM measurement) is performed onto message *codewords*, defined by $\sigma_M = \sigma_{m_1} \otimes \sigma_{m_2} \otimes \cdots \otimes \sigma_{m_n}$,

given the originator codeword $\rho_M = \rho_{m_1} \otimes \rho_{m_2} \otimes \cdots \otimes \rho_{m_n}$. In the received message σ_M , each symbol σ_{m_k} is the result of a quantum operation ε such that $\sigma_{m_k} = \varepsilon(\rho_{m_k})$. From statistical measurements, the recipient may then compute $\chi' = S(\langle\sigma\rangle) - \langle S(\sigma) \rangle = S[\langle\varepsilon(\rho)\rangle] - \langle S[\varepsilon(\rho)] \rangle$ for the source σ , but it is irrelevant to call this quantity a “Holevo bound,” because of the quantum operation ε introduced by the channel, which, for each recipient symbol σ_{m_k} , has corrupted the reference symbol ρ_{m_k} . If ε is not a “noiseless” constant, therefore, the action of decoding is conceptually quite different from that assumed in the Holevo bound derivation. Furthermore, however tempting, we ought not to view χ' as representing some elaborated definition of “quantum mutual information” between the sources ρ and σ of two quantum systems Q (originator) and Q' (recipient). In Chapter 21, we have described the *quantum mutual information* $S(\hat{\rho}_A; \hat{\rho}_B)$ for a composite system AB as

$$S(\hat{\rho}_A; \hat{\rho}_B) = S(\hat{\rho}_A) + S(\hat{\rho}_B) - S(\rho_{AB}), \quad (23.44)$$

where $\hat{\rho}_A = \text{tr}_B(\rho_{AB})$ and $\hat{\rho}_B = \text{tr}_A(\rho_{AB})$ are the partial-traced operators. Life in QIT would be so much simpler if $\max[S(\hat{\rho}_Q; \hat{\rho}_{Q'})]$ could represent the quantum channel capacity χ , but this is definitely not the case! To cut through any other explanation, it is not possible to define the composite system QQ' with density operator $\rho_{QQ'}$, simply because after the operation ε the original system Q does not exist anymore as it has been transformed into Q' ! The physical parameter that can make sense is the difference $\Delta S = S(\rho_{Q'}) - S(\rho_Q) \equiv S(\sigma) - S(\rho)$, see further on.

The quantum counterpart of mutual information is, indeed, far more complex, and its description is beyond the scope of this chapter. Here, to quench our thirst, I may provide just a brief hint of a new concept referred to as *quantum coherent information*. Appendix W shows that given a quantum system Q in a mixed state $|\psi_Q\rangle$ with density operator $\rho_Q \equiv \rho$, it is possible to define a reference quantum system R , such that the composite system QR is in a pure state $|\psi_{QR}\rangle$, namely, for which $\rho_{QR} = |\psi_{QR}\rangle\langle\psi_{QR}|$. Such an operation is referred to as *state purification*. Upon the action of the quantum operation ε , the systems Q, R are then transformed into the systems Q', R' with, in particular, $\rho_{Q'} \equiv \sigma = \varepsilon(\rho)$. One defines the *entropy exchange* of the action of ε on the composite system QR that transforms it into the composite system $Q'R'$ as

$$S_E(\rho, \varepsilon) = S(\rho_{Q'R'}). \quad (23.45)$$

Then one defines the *quantum coherent information* as the difference

$$I(\rho, \varepsilon) = S(\sigma) - S_E(\rho, \varepsilon). \quad (23.46)$$

In today’s state of the art of QIT, the quantum coherent information $I(\rho, \varepsilon)$ is only “conjectured” as playing the same role as the mutual information

$$H(X; Y) = H(Y) - H(Y|X) \quad (23.47)$$

does in Shannon’s classical theory. In particular, it can be shown that given the entropy difference between source and originator,

$$\Delta S = S(\sigma) - S(\rho), \quad (23.48)$$

the following property is always satisfied:

$$\Delta S + S_E(\rho, \varepsilon) \geq 0. \quad (23.49)$$

This last result shows that the entropy change caused by the quantum operation ε on both the quantum channel and the “outside world” is always nonnegative, which recalls the *second law of thermodynamics*, according to which the entropy of a closed system may only increase on any physical transformation or evolution from its initial state.

23.3 Capacity of some quantum channels

In this section, I apply the HSW theorem to evaluate the capacity of different types of quantum channel. We assume that the originator source has, for symbols, the pure states $\rho_i = |\psi_i\rangle\langle\psi_i|$ which are associated within a quantum message (or codeword) with some probability distribution p_i . For convenience, we introduce

$$\chi' = S \left[\varepsilon \left(\sum_i p_i \rho_i \right) \right] - \sum_i p_i S[\varepsilon(\rho_i)], \quad (23.50)$$

to which, after the HSW theorem, the channel capacity $\chi = \max(\chi')$ corresponds.

Before touching upon any examples, it is interesting to consider the case of ideal, constant or “noiseless” channels. This is the case where $\varepsilon(\rho_i) = \rho_i$ and, hence, from Eq. (23.50):

$$\chi'_{\text{ideal}} = \left(\sum_i p_i \rho_i \right) - \sum_i p_i S(\rho_i) \equiv S \left(\sum_i p_i \rho_i \right), \quad (23.51)$$

where we used the property that for any pure state ρ_i the VN entropy $S(\rho_i)$ is zero. The result in Eq. (23.49) shows that in this ideal case, χ' actually represents the VN entropy of the originator’s message source $S(\rho) = S(\langle \rho_i \rangle)$. The “ideal channel” capacity is given by $\chi_{\text{ideal}} = \max(\chi'_{\text{ideal}})$, or

$$\chi_{\text{ideal}} = \max(\chi'_{\text{ideal}}) = \max_{p_i} \left[S \left(\sum_i p_i \rho_i \right) \right] \equiv \max S(\rho). \quad (23.52)$$

This maximization problem, thus, addresses the question of how much classical information can be conveyed through any given message ρ before transmission through any quantum channel. Intuitively, we expect that for nonideal channels we have the property

$$\chi \leq \chi_{\text{ideal}}, \quad (23.53)$$

meaning that, obviously, *noisy transmission does not improve the information of the message source*. This is just like in the classical case, with the mutual information satisfying $H(X; Y) \leq H(X)$, and with the capacity $C = \max[H(X; Y)] \leq \max[H(X)] = 1$ bit. The following examples will illustrate the fact that the quantum channel capacity χ cannot, indeed, exceed the highest-possible VN entropy of the originator source, i.e., $\chi_{\text{ideal}} = \max[S(\rho)]$.

Example 23.1: Depolarizing channel

In Eq. (23.8) we have seen that this channel is defined by the operation

$$\varepsilon(\rho_i) = p \frac{I}{2} + (1-p)\rho_i. \quad (23.54)$$

Assuming the orthogonal symbols $\rho_1 = |0\rangle\langle 0|$ and $\rho_2 = |1\rangle\langle 1|$, we obtain

$$\begin{aligned} \varepsilon\left(\sum_i p_i \rho_i\right) &= \varepsilon[p_1 \rho_1 + (1-p_1)\rho_2] \\ &= p \frac{I}{2} + (1-p)[p_1 \rho_1 + (1-p_1)\rho_2] \\ &= \frac{p}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + (1-p)p_1 \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + (1-p)(1-p_1) \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \\ &= \frac{p}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + (1-p) \begin{pmatrix} p_1 & 0 \\ 0 & 1-p_1 \end{pmatrix} \\ &\equiv \begin{pmatrix} \frac{p}{2} + (1-p)p_1 & 0 \\ 0 & \frac{p}{2} + (1-p)(1-p_1) \end{pmatrix}, \end{aligned} \quad (23.55)$$

and the corresponding VN entropy

$$S\left[\varepsilon\left(\sum_i p_i \rho_i\right)\right] = - \left[\frac{\frac{p}{2} + 2(1-p)p_1}{2} \log \frac{\frac{p}{2} + 2(1-p)p_1}{2} + \frac{\frac{p}{2} + 2(1-p)(1-p_1)}{2} \log \frac{\frac{p}{2} + 2(1-p)(1-p_1)}{2} \right]. \quad (23.56)$$

As we have seen in the previous section, the VN entropy of the recipient symbols is $S[\varepsilon(\rho_1)] = S[\varepsilon(\rho_2)] = f(p/2)$, hence,

$$\sum_i p_i S[\varepsilon(\rho_i)] = p_1 f\left(\frac{p}{2}\right) + (1-p_1) f\left(\frac{p}{2}\right) \equiv f\left(\frac{p}{2}\right). \quad (23.57)$$

From Eq. (23.50), we finally obtain

$$\begin{aligned} \chi'(p, p_1) &= S\left[\varepsilon\left(\sum_i p_i \rho_i\right)\right] - \sum_i p_i S[\varepsilon(\rho_i)] \\ &= - \left\{ \left[\frac{p}{2} + (1-p)p_1 \right] \log \left[\frac{p}{2} + (1-p)p_1 \right] + \left[\frac{p}{2} + (1-p)(1-p_1) \right] \log \left[\frac{p}{2} + (1-p)(1-p_1) \right] \right\} - f\left(\frac{p}{2}\right). \end{aligned} \quad (23.58)$$

In the above equation, it is clear that the first term in braces, corresponding to the definition in Eq. (23.56), is maximized if we choose the two symbols to have equal probabilities $p_1 = 1 - p_2 = 1/2$, which gives

$$S\left[\varepsilon\left(\sum_i p_i \rho_i\right)\right] = - \left(\frac{1}{2} \log \frac{1}{2} + \frac{1}{2} \log \frac{1}{2} \right) = f\left(\frac{1}{2}\right) = 1. \quad (23.59)$$

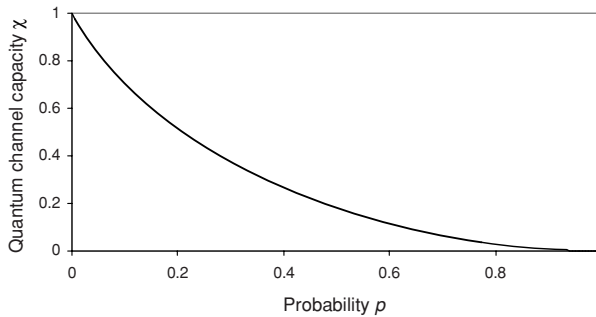


Figure 23.1 Quantum capacity $\chi(p)$ of the depolarizing channel as a function of the depolarizing parameter p .

Thus, we obtain the channel capacity χ :

$$\chi(p) = \max \chi' \left(p, \frac{1}{2} \right) \equiv 1 - f \left(\frac{p}{2} \right). \quad (23.60)$$

As expected from the analysis in the previous section, the capacity vanishes for $p = 1$, which corresponds to a “useless channel,” projecting any random symbol ρ_i onto $\sigma = \varepsilon(\rho_i) = I/2$ as shown from Eq. (23.54). The highest capacity $\chi(0) = 1$ bit is obtained for $p = 0$, in which case the quantum channel ε is constant or “ideal,” or “noiseless,” i.e., $\sigma = \varepsilon(\rho_i) = \rho_i$, but this case is of no interest here. Figure 23.1 shows a plot of the depolarizing channel capacity as a function of the depolarizing probability parameter p . It is seen from the figure that, as expected, the channel capacity monotonously decays as the depolarizing parameter increases from $p = 0$ to $p = 1$. To provide a practical engineering interpretation of the above result, assume, for instance, that $p = 0.1$, which yields $f(p/2) \approx 0.25$ (see Fig. 4.7), and, thus, defines a channel capacity of $\chi = 0.75$ bit. The HSW theorem states that it is possible to transmit code-words of sufficient lengths with arbitrary error probability provided the code rate satisfies $R < \chi = 0.75$. Given a block code (n, k) , the code rate must satisfy $R = k/n < 0.75$, which means that the code must include 25% redundancy bits and 75% payload bits.

The exercises provide other application examples of the depolarizing channel capacity. Here, it is worth showing the results obtained when assuming the codeword symbols $|0\rangle, |+\rangle$ instead of $|0\rangle, |1\rangle$. It is a tedious but tractable exercise to show that the corresponding capacity takes the form:

$$\chi(p) = \max_{p_1} \left\{ \begin{aligned} & -\frac{1 + (1-p)\sqrt{1-2p_1(1-p_1)}}{2} \log \frac{1 + (1-p)\sqrt{1-2p_1(1-p_1)}}{2} \\ & -\frac{1 - (1-p)\sqrt{1-2p_1(1-p_1)}}{2} \log \frac{1 - (1-p)\sqrt{1-2p_1(1-p_1)}}{2} \\ & + (1-p_1) \frac{1 + \sqrt{1-p(2-p)}}{2} \log \frac{1 + \sqrt{1-p(2-p)}}{2} \\ & + (1-p_1) \frac{1 - \sqrt{1-p(2-p)}}{2} \log \frac{1 - \sqrt{1-p(2-p)}}{2} - p_1 f \left(\frac{p}{2} \right) \end{aligned} \right\}. \quad (23.61)$$

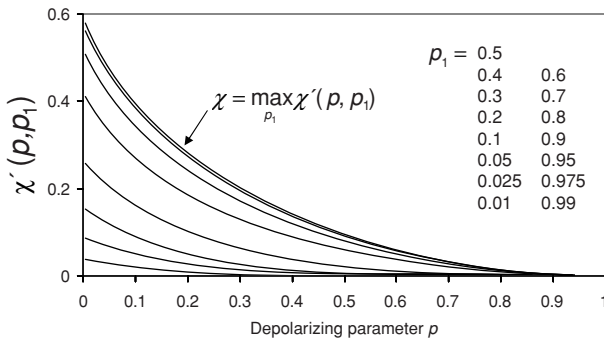


Figure 23.2 Plots of $\chi'(p, p_1)$ as a function of the depolarizing parameter p in depolarizing-channel, and corresponding channel capacity $\chi(p) = \max \chi'(p, p_1)$ (codewords formed by pure states $|0\rangle, |+\rangle$ with probabilities $p_1, 1 - p_1$, respectively).

The expression in brackets $\{\chi'(p, p_1)\}$ in Eq. (23.61) can be maximized by numerically solving the transcendental equation $\partial \chi'(p, p_1) / \partial p_1 = 0$. But we may as well directly infer the answer, as it will be shown further on. First, we may just plot the function $\chi'(p, p_1)$ for different values of the parameter p_1 (probability associated with $|0\rangle$), as illustrated in Fig. 23.2. From the figure, we first note the property $\chi'(p, p_1) = \chi'(p, 1 - p_1)$. Second, we observe that in all cases, the function $\chi'(p, p_1)$ is maximal at $p = 0$ (constant or noiseless channel) and zero at $p = 1$ (useless channel). Third, we see that the channel capacity $\chi(p) = \max \chi'(p, p_1)$ is achieved for $p_1 = 0.5$. This was expected, since this condition corresponds to *maximum uncertainty* in the occurrence of the $|0\rangle, |+\rangle$ qubits forming the codewords. From the property in Eq. (23.53), the constant or noiseless capacity, $\chi(0)$, must correspond to the (maximum possible) VN entropy of originator's source, $\max[S(\rho)] = \chi_{\text{ideal}}$, as discussed earlier and also formally established in the exercise. Substituting $p_1 = 0.5$ into Eq. (23.61), we obtain the following analytical form of the channel capacity:

$$\chi \equiv \chi' \left(p, \frac{1}{2} \right) = \left\{ \begin{array}{l} - \frac{1 + \frac{1-p}{\sqrt{2}}}{2} \log \frac{1 + \frac{1-p}{\sqrt{2}}}{2} \\ - \frac{1 - \frac{1-p}{\sqrt{2}}}{2} \log \frac{1 - \frac{1-p}{\sqrt{2}}}{2} \\ + \frac{1 + \sqrt{1-p(2-p)}}{4} \log \frac{1 + \sqrt{1-p(2-p)}}{2} \\ + \frac{1 - \sqrt{1-p(2-p)}}{4} \log \frac{1 - \sqrt{1-p(2-p)}}{2} - \frac{1}{2} f \left(\frac{p}{2} \right) \end{array} \right\}. \quad (23.62)$$

Substituting $p = 0$ in the above, we get the upper bound for the channel capacity, which corresponds to the originator's maximum source entropy $\max[S(\rho)]$:

$$\begin{aligned}\chi_{\text{ideal}} &= \max S(\rho) \\ &= \chi' \left(0, \frac{1}{2} \right) \\ &\equiv - \left\{ \frac{2 + \sqrt{2}}{4} \log \frac{2 + \sqrt{2}}{4} + \frac{2 - \sqrt{2}}{4} \log \frac{2 - \sqrt{2}}{4} \right\}.\end{aligned}\quad (23.63)$$

This example, thus, provides an illustration of the HSW maximization problem, leading, in this case, to a tractable analytical solution. It also illustrates the property $\chi(p) \leq \chi_{\text{ideal}} = \max S(\rho)$, which can be used to guide the maximization problem solving.

Example 23.2: Bit-flip channel

The quantum operation of this channel is defined in Eq. (23.13). Assuming the orthogonal symbols $\rho_1 = |0\rangle\langle 0|$ and $\rho_2 = |1\rangle\langle 1|$, we obtain

$$\begin{aligned}\varepsilon \left(\sum_i p_i \rho_i \right) &= \varepsilon [p_1 \rho_1 + (1 - p_1) \rho_2] \\ &= pX[p_1 \rho_1 + (1 - p_1) \rho_2]X + (1 - p)[p_1 \rho_1 + (1 - p_1) \rho_2] \\ &= pp_1 X \rho_1 X + p(1 - p_1) X \rho_2 X + (1 - p)p_1 \rho_1 + (1 - p)(1 - p_1) \rho_2 \\ &= \left[pp_1 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + p(1 - p_1) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right] \\ &\quad + (1 - p)p_1 \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + (1 - p)(1 - p_1) \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \\ &= pp_1 \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} + p(1 - p_1) \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} (1 - p)p_1 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & (1 - p)(1 - p_1) \end{pmatrix} \\ &\equiv \begin{pmatrix} p + p_1 - 2pp_1 & 0 \\ 0 & 1 - [p + p_1 - 2pp_1] \end{pmatrix}.\end{aligned}\quad (23.64)$$

and the corresponding VN entropy

$$\begin{aligned}A &= S \left[\varepsilon \left(\sum_i p_i \rho_i \right) \right] \\ &= - \left\{ (p + p_1 - 2pp_1) \log (p + p_1 - 2pp_1) \right. \\ &\quad \left. + [1 - (p + p_1 - 2pp_1)] \log [1 - (p + p_1 - 2pp_1)] \right\} \\ &\equiv f(p + p_1 - 2pp_1).\end{aligned}\quad (23.65)$$

Next, we calculate the quantum operation ε onto ρ_1 and ρ_2 along with the corresponding entropies:

$$\begin{aligned}\varepsilon(\rho_1) &= pX\rho_1X + (1-p)\rho_1 \\ &\equiv p \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} + (1-p) \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \\ &\equiv \begin{pmatrix} 1-p & 0 \\ 0 & p \end{pmatrix},\end{aligned}\tag{23.66}$$

$$\begin{aligned}\rightarrow \\ S[\varepsilon(\rho_1)] &= -[(1-p)\log(1-p) + p\log p] \\ &\equiv f(1-p) \\ &\equiv f(p)\end{aligned}$$

$$\begin{aligned}\varepsilon(\rho_2) &= pX\rho_2X + (1-p)\rho_2 \\ &\equiv p \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + (1-p) \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \\ &\equiv \begin{pmatrix} p & 0 \\ 0 & 1-p \end{pmatrix}\end{aligned}\tag{23.67}$$

$$\begin{aligned}\rightarrow \\ S[\varepsilon(\rho_2)] &\equiv f(p).\end{aligned}$$

Hence, we obtain

$$\begin{aligned}B &= \sum_i p_i S[\varepsilon(\rho_i)] \\ &= p_1 f(p) + (1-p_1) f(p) \\ &\equiv f(p).\end{aligned}\tag{23.68}$$

Combining the results in Eqs. (23.65) and (23.68), we finally obtain the channel capacity

$$\begin{aligned}\chi(p, p_1) &= \max_{p_1}(\chi') \\ &\equiv \max_{p_1}(A - B) \\ &\equiv \max_{p_1}[f(p + p_1 - 2pp_1) - f(p)].\end{aligned}\tag{23.69}$$

We note that the argument satisfies the property $\chi'(p, p_1) = \chi'(p, 1 - p_1)$. The maximization problem can be solved analytically by studying the partial derivatives $\partial\chi'(p, p_1)/\partial p_1$ and $\partial^2\chi'(p, p_1)/\partial p_1^2$, but we may observe, again, that the capacity cannot exceed the amount $\chi_{\text{ideal}} = \max S(\rho)$. Using the definition of the symbol density operator $\rho = p_1\rho_1 + (1 - p_1)\rho_2$, we clearly have $S(\rho) = f(p_1)$, and $\chi_{\text{ideal}} = \max S(\rho)$, yielding $p_1 = 1/2$, as expected (maximum VN entropy is obtained for a uniform message-symbol distribution). Thus, we have for all p the channel capacity:

$$\begin{aligned}\chi(p) &= \chi'\left(p, \frac{1}{2}\right) \\ &= [f(p + 1/2 - 2p/2) - f(p)] \\ &\equiv 1 - f(p).\end{aligned}\tag{23.70}$$

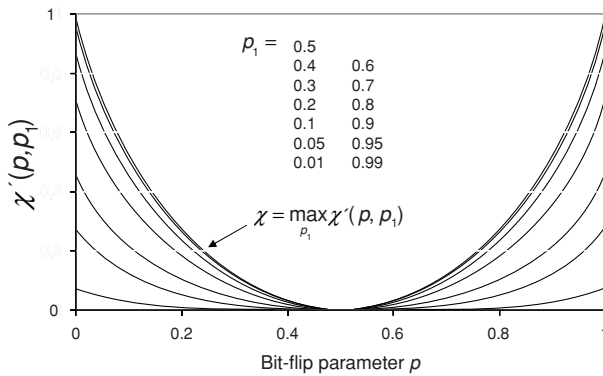


Figure 23.3 Plots of $\chi'(p, p_1)$ as a function of the bit-flipping parameter p in the bit-flip channel, and corresponding channel capacity $\chi(p) = \max \chi'(p, p_1)$ (codewords formed by pure states $|0\rangle, |+\rangle$ with probabilities $p_1, 1 - p_1$, respectively).

Figure 23.3 shows plots of $\chi'(p, p_1)$ for $p_1 = 0$ to $p_1 = 1$ through different increments (recalling that $\chi'(p, p_1) = \chi'(p, 1 - p_1)$), with the top curve $\chi(p) = \chi'(p, 1/2)$ representing the channel capacity. Interestingly, the channel capacity is seen to have two maxima, i.e., for $p = 0$ and $p = 1$. These two limiting cases correspond to the noiseless channel, and the “deterministic” bit-flip channel, respectively. A deterministic bit-flipping is simply a change of code polarity, meaning that the classical cbit codewords from originator to recipient are exactly inverted or complemented, which entails no information degradation or error. The other limiting situation is obtained for $p = 1/2$, meaning that the qubit has a 50% chance of being flipped and a 50% chance of being conserved in its integrity. The recipient’s measurement amounts to a coin-flipping experiment, and all initial information is lost, which is the situation of the *useless channel*.

23.4 Exercises

23.1 (B): Given the Pauli matrices X, Y, Z (see definition in Chapter 16), show that for any 2×2 operator A with unity trace, and for $q = 3p/4$, the following relation holds:

$$\frac{q}{3}(XAX + YAY + ZAZ) + (1 - q)A = p\frac{I}{2} + (1 - p)A.$$

23.2 (M): Analyze the effect of the bit-flip quantum channel defined by

$$\sigma = \varepsilon(\rho) = pX\rho X + (1 - p)\rho,$$

with p being a probability distribution and X the Pauli matrix

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

in the case where the density operator ρ is nondiagonal. Determine the entropy change $\Delta S = S(\sigma) - S(\rho)$ in the case $\rho = |0\rangle\langle 0|$ and $\rho = |1\rangle\langle 1|$.

23.3 (M): Analyze the effect of the phase-flip quantum channel as defined by

$$\sigma = \varepsilon(\rho) = pZ\rho Z + (1 - p)\rho,$$

with p being a probability distribution and Z the Pauli matrix

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

in the case where the density operator ρ is nondiagonal. Determine the entropy change $\Delta S = S(\sigma) - S(\rho)$ in the four cases $\rho = |0\rangle\langle 0|$, $\rho = |1\rangle\langle 1|$, $\rho = |+\rangle\langle +|$ and $\rho = |-\rangle\langle -|$.

23.4 (M): Analyze the effect of the bit-phase-flip quantum channel as defined by

$$\sigma = \varepsilon(\rho) = pY\rho Y + (1 - p)\rho,$$

with p being a probability distribution and Y the Pauli matrix

$$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix},$$

in the case where the density operator ρ is nondiagonal. Determine the entropy change $\Delta S = S(\sigma) - S(\rho)$ in the four cases $\rho = |0\rangle\langle 0|$, $\rho = |1\rangle\langle 1|$, $\rho = |+\rangle\langle +|$, and $\rho = |-\rangle\langle -|$.

23.5 (M): Determine the quantum channel capacity corresponding to the depolarizing channel

$$\varepsilon(\rho_i) = p \frac{I}{3} + (1 - p)\rho_i,$$

assuming the three-symbol alphabet

$$\rho_1 = |0\rangle\langle 0|, \rho_2 = |1\rangle\langle 1|, \rho_3 = |2\rangle\langle 2|$$

associated with probabilities p_1, p_2, p_3 . Provide a plot of the capacity as a function of the depolarizing parameter p .

23.6 (T): Determine the quantum channel capacity corresponding to the depolarizing channel

$$\varepsilon(\rho) = pZ\rho Z + (1 - p)\rho,$$

assuming the two-symbol alphabet

$$\rho_1 = |0\rangle\langle 0|, \rho_2 = |+\rangle\langle +| = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

associated with probabilities p_1, p_2 .

Hint: plot

$$\chi'(p) = S(\langle \varepsilon \rangle) - \langle S(\varepsilon) \rangle$$

as a function of p , for different values of p_1 and justify that the capacity $\chi = \max(\chi')$ is achieved for $p_1 = p_2 = 1/2$; also show that $\chi(p) \leq \chi(0) = S(\rho)$, where $\chi(0)$ is the capacity of the constant or noiseless quantum channel and $S(\rho)$ is the VN entropy of the originator source.